

Project Executable Files

Date	20 July 2026
Project Name	OptiCrop: Smart Agricultural Production Optimization Engine

Step 1: Submission Checklist

Step 4: Run Instructions

```
cd "5. Project Development Phase"
pip install -r requirements.txt
python train_model.py # optional - regenerates model.pkl and EDA plots
python app.py
```

Open `http://127.0.0.1:5000` in a browser, navigate to **Find Your Crop**, enter soil (N, P, K, pH) and climate (temperature, humidity, rainfall) values, and click **Predict Best Crop**. For a production-style deployment, the included `Procfile` (`web: gunicorn app:app`) is ready to use on platforms like Heroku.

Step 5: Known Issues / Limitations

S.No	Known Issue / Limitation	Workaround / Status
1	Only shows a single top prediction, not top-3 alternatives	Planned for future roadmap
2	No persistent storage of past predictions	By design — stateless single-session app
3	Flask dev server not production-hardened	Use Gunicorn/WSGI server for deployment